

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – MODULE 4

Database Management – SQL & PL/SQL

SESSION / ASSIGNMENT – 3

TASKS :

1. Create a SQL table called Restaurants with columns: id (INT, auto-increment), name (VARCHAR), cuisine (VARCHAR), rating (DECIMAL), and city (VARCHAR). Insert 5 sample restaurants into the table, each with a different cuisine and rating.

→ CODE:

```
CREATE TABLE Restaurants (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    name VARCHAR(100),  
    cuisine VARCHAR(50),  
    rating DECIMAL(2,1),  
    city VARCHAR(50)  
);
```

```
INSERT INTO Restaurants (name, cuisine, rating, city)  
VALUES  
( 'Spice Villa', 'Indian', 4.5, 'Mumbai'),  
( 'Sushi World', 'Japanese', 4.2, 'Delhi'),  
( 'La Bella Italia', 'Italian', 4.7, 'Ahmedabad'),  
( 'Taco Fiesta', 'Mexican', 4.1, 'Surat'),  
( 'Dragon Palace', 'Chinese', 4.4, 'Vadodara');
```

```
SELECT * FROM playlists;
```

OUTPUT:

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0004 seconds.)

```
CREATE TABLE Restaurants ( id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(100), cuisine VARCHAR(50), rating DECIMAL(2,1), city VARCHAR(50) );
```

[Edit inline] [Edit] [Create PHP code]

✓ 5 rows inserted.

Inserted row id: 5 (Query took 0.0009 seconds.)

```
INSERT INTO Restaurants (name, cuisine, rating, city) VALUES ('The Spice House', 'Indian', 4.5, 'Mumbai'), ('Sushi World', 'Japanese', 4.2, 'Delhi'), ('La Bella Italia', 'Italian', 4.7, 'Ahmedabad'), ('Taco Fiesta', 'Mexican', 4.1, 'Surat'), ('Dragon Palace', 'Chinese', 4.4, 'Vadodara');
```

[Edit inline] [Edit] [Create PHP code]

✓ MySQL returned an empty result set (i.e. zero rows). (Query took 0.0001 seconds.)

```
SELECT * FROM playlists;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

←T→				id	name	cuisine	rating	city
☐	 Edit	 Copy	 Delete	1	Spice Villa	Indian	4.5	Mumbai
☐	 Edit	 Copy	 Delete	2	Sushi World	Japanese	4.2	Delhi
☐	 Edit	 Copy	 Delete	3	La Bella Italia	Italian	4.7	Ahmedabad
☐	 Edit	 Copy	 Delete	4	Taco Fiesta	Mexican	4.1	Surat
☐	 Edit	 Copy	 Delete	5	Dragon Palace	Chinese	4.4	Vadodara

2. Write a SQL UPDATE statement to change the rating of the restaurant named 'Spice Villa' to 4.7 in the Restaurants table.

→ CODE:

```
UPDATE restaurants
```

```
SET rating = 4.7
```

```
WHERE name = 'Spice Villa';
```

```
SELECT * FROM restaurants;
```

OUTPUT:

				id	name	cuisine	rating	city
☐	 Edit	 Copy	 Delete	1	Spice Villa	Indian	4.7	Mumbai
☐	 Edit	 Copy	 Delete	2	Sushi World	Japanese	4.2	Delhi
☐	 Edit	 Copy	 Delete	3	La Bella Italia	Italian	4.7	Ahmedabad
☐	 Edit	 Copy	 Delete	4	Taco Fiesta	Mexican	4.1	Surat
☐	 Edit	 Copy	 Delete	5	Dragon Palace	Chinese	4.4	Vadodara

3. Delete all restaurants from the Restaurants table where the rating is less than 3.5 using a SQL DELETE statement.

➔ CODE:

```
DELETE FROM Restaurants
WHERE rating < 3.5;
```

```
SELECT * FROM Restaurants;
```

OUTPUT:

✓ 2 rows deleted. (Query took 0.0006 seconds.)

```
DELETE FROM Restaurants WHERE rating < 3.5;
```

[Edit inline] [Edit] [Create PHP code]

✓ Showing rows 0 - 2 (3 total, Query took 0.0004 seconds.)

```
SELECT * FROM Restaurants;
```

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 ▼ Filter rows:

Extra options

				id	name	cuisine	rating	city			
☐		Edit		Copy		Delete	1	Spice Villa	Indian	4.7	Mumbai
☐		Edit		Copy		Delete	3	La Bella Italia	Italian	4.7	Ahmedabad
☐		Edit		Copy		Delete	4	Taco Fiesta	Mexican	4.1	Surat

← T →						id	name	cuisine	rating	city
☐	 Edit	 Copy	 Delete	1	Spice Villa	Indian	4.7	Mumbai		
☐	 Edit	 Copy	 Delete	3	La Bella Italia	Italian	4.7	Ahmedabad		
☐	 Edit	 Copy	 Delete	4	Taco Fiesta	Mexican	4.1	Surat		

4. Use a **SELECT** query with **WHERE** and **LIMIT** to fetch the top 2 restaurants in the city 'Ahmedabad' with a rating above 4.0 from the Restaurants table.

Hint: Combine **WHERE** for city and rating, and **LIMIT** to restrict the number of results.

➔ **CODE:**

```
SELECT * FROM Restaurants
WHERE city = 'Ahmedabad'
AND rating > 4.0
LIMIT 2;
```

OUTPUT:

✓ Showing rows 0 - 0 (1 total, Query took 0.0003 seconds.)

```
SELECT * FROM Restaurants WHERE city = 'Ahmedabad' AND rating > 4.0 LIMIT 2;
```

☐ Profiling [[Edit inline](#)] [[Edit](#)] [[Explain SQL](#)] [[Create PHP code](#)] [[Refresh](#)]

Extra options

← T →

▼

	id	name	cuisine	rating	city
☐ Edit Copy Delete	3	La Bella Italia	Italian	4.7	Ahmedabad